Effects of Intermittent Fasting on Lymphocytes, Quality of Life, and Weight

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Hypotheses

A Study of Intermittent Fasting (**IF**)
in Chronic Lymphocytic Leukemia and Small Lymphocytic Lymphoma (CLL/SLL) Patients at BC Cancer - Victoria

IF can help patients

- lower their lymphocyte count (`LYMP`);
- improve their health score (`QoL`);
- lose body weight.



Image: Atkins

Data

Repeated-measures (within-subject) data are usually (1) longitudinal or (2) clustered. Measurements on (1) the same subjects over time or (2) all subjects within a family are **correlated**.

A total of 15 subjects in three groups:

- three-month fasting and one-month follow-up,

  IF-002, 004, 005, 006, and 007;

- six-month fasting,

  IF-017, 019, 020, 021, and 022;

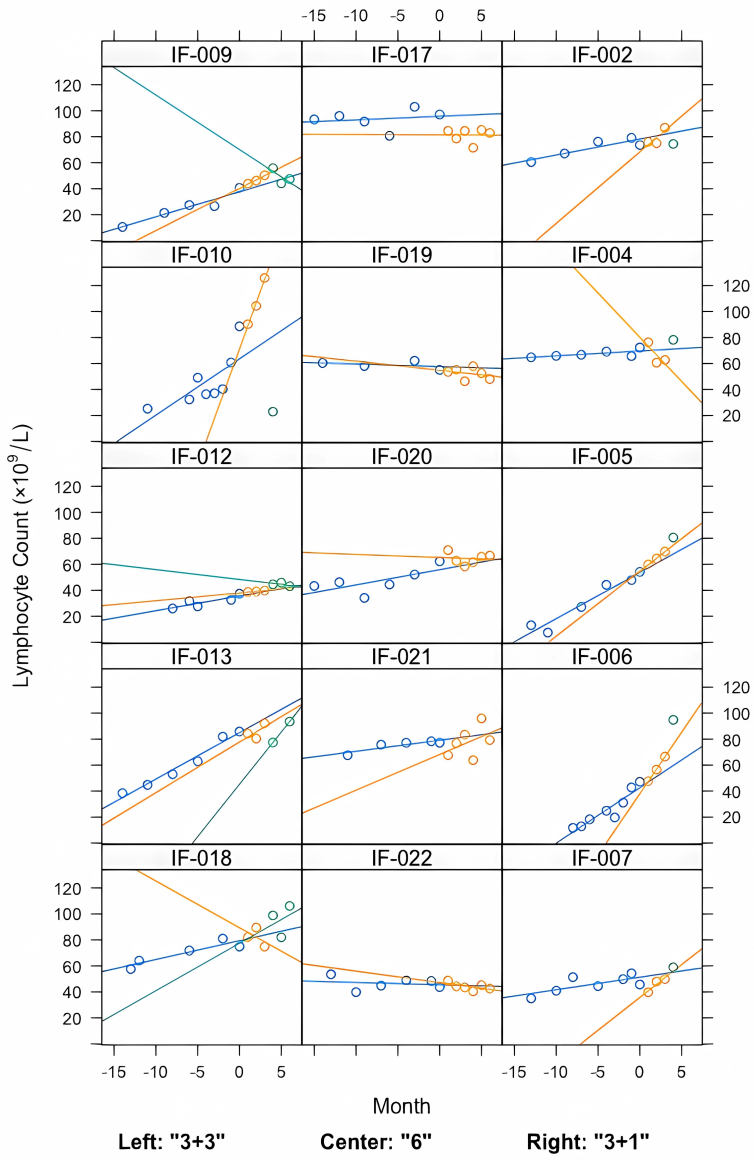
- three-month fasting and three-month follow-up,

  IF-009, 010, 012, 013, and 018.

IF_v3.1.csv

```
> head(df, 9)
    LYMP  ATG      ID Time Time_Within_Phase Phase
1  60.5   NA IF-002  -13                 0     0
2  67.1   NA IF-002   -9                 4     0
3  76.2   NA IF-002   -5                 8     0
4  79.1   NA IF-002   -1                12     0
5  73.7 0.31 IF-002    0                13     0
6  75.9 0.20 IF-002    1                 0     1
7  75.0   NA IF-002    2                 1     1
8  86.9 0.22 IF-002    3                 2     1
9  74.4 0.20 IF-002    4                 0     2
```

```
> tail(df, 9)
    LYMP  ATG      ID Time Time_Within_Phase Phase
158 48.9   NA IF-022   -4                 9     0
159 48.4   NA IF-022   -1                12     0
160 43.7 0.95 IF-022    0                13     0
161 48.7 0.59 IF-022    1                 0     1
162 44.3 1.00 IF-022    2                 1     1
163 43.5 0.68 IF-022    3                 2     1
164 40.4 0.64 IF-022    4                 3     1
165 45.2 0.90 IF-022    5                 4     1
166 42.6   NA IF-022    6                 5     1
```



Methods[1]: LYMP

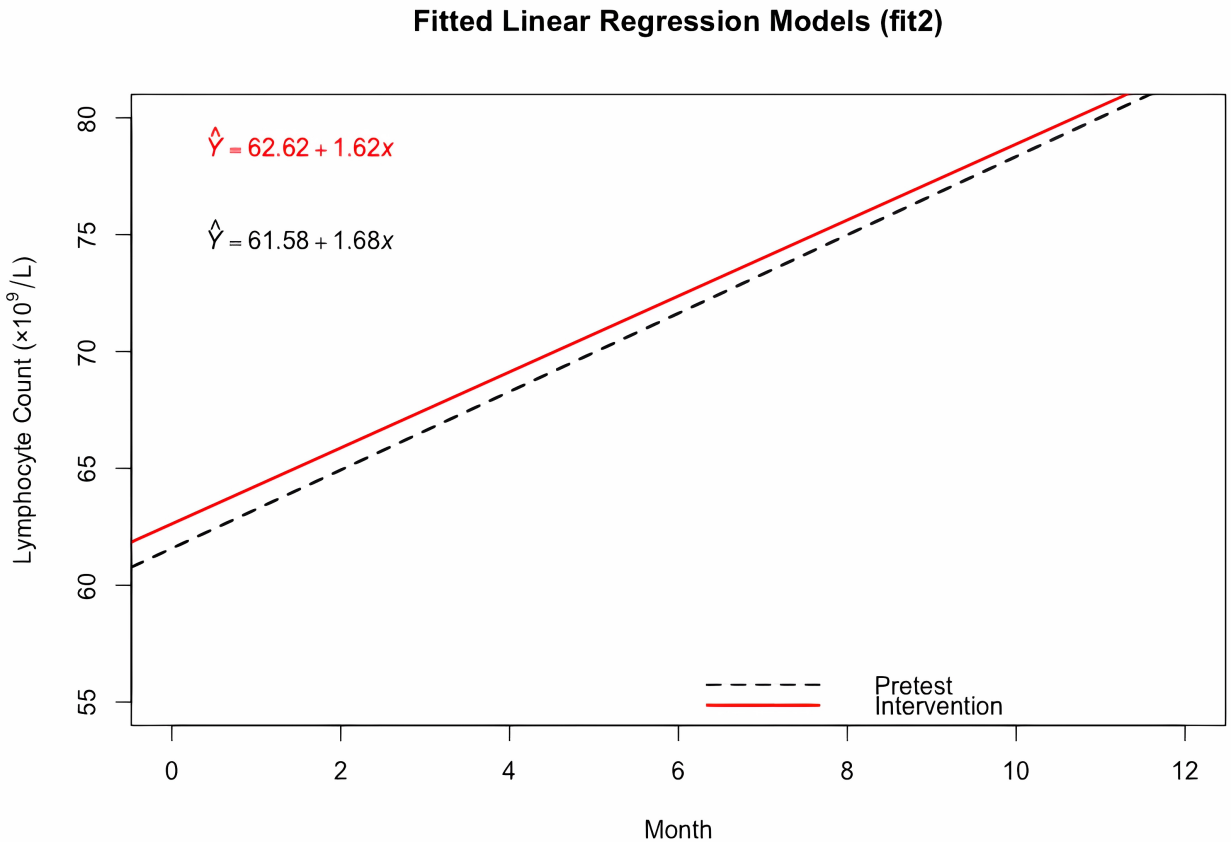
In a linear mixed-effects regression, each subject j can have its own intercept and slope over time.

$$Y = \underbrace{(\beta_0}_{\text{fixed intercept}} + \underbrace{u_{0j})}_{\text{random intercept}} + \underbrace{(\beta_1}_{\text{fixed slope}} + \underbrace{u_{1j})}_{\text{random slope}} \cdot \text{Time}$$
$$+(\beta_2 + u_{2j}) \cdot \text{Phase} + (\beta_3 + u_{3j}) \cdot [\text{Phase} \times \text{Time}] + \epsilon \qquad (1)$$

```
> coef(fit2)
       (Intercept) factor(Phase)1        Time factor(Phase)1:Time
IF-002    76.64484         1.0489  1.12003288         -0.05163969
IF-004    66.97597         1.0489  0.09173338         -0.05163969
IF-005    55.16979         1.0489  3.66212044         -0.05163969
...

> coef(fit2u)
       (Intercept) factor(Phase)1        Time factor(Phase)1:Time
IF-002    78.73166     -4.2850300   1.3723616           0.9039365
IF-004    70.96807     -5.6374723   0.5528317          -0.5026841
IF-005    49.98457      3.7726349   3.0038973           3.1135593
...
```

Results[1]: LYMP



The mean lymphocyte count continued to increase at the same rate after participants received IF treatment. (cf. *IF v3.7.Rmd* p. 3-5)

($p = .94$, $t_{130} = -0.07$, 95% CI $= [-1.49,\ 1.39]$, and $BF_{01} = 2.69$)

Results[1] (continued): LYMP

Change `Time` to `Time_Within_Phase` and `Phase` to `Fast`. Subset `Phase==1`.

Participants who fasted for three months showed a higher lymphocyte growth rate than those who fasted for six months during the intervention. (cf. *IF v3.7.Rmd* p. 9)

($p = .047$, $t_{43} = 2.05$, 95% CI = $[0.18, 7.49]$, and $BF_{01} = 0.98$)

There was no difference between the autophagy values of fasting for three and six months. (cf. *IF v3.7.Rmd* p. 11)

($p = .20$, $t_{34} = -1.31$, 95% CI = $[-0.22, 0.04]$, and $BF_{01} = 1.85$)

Methods[2]: `LYMP`

If a change point exists at t by some criterion,
then $\{Y_1, \overset{\text{i.i.d.}}{\cdots}, Y_t\}$ will differ from $\{Y_{t+1}, \overset{\text{i.i.d.}}{\cdots}, Y_n\}$ in their means, variances, distributions, or some other parameters of interest.



changepoint – changes in mean and variance

strucchange – changes in regression

changepoint.np – nonparametric

bcp – Bayesian

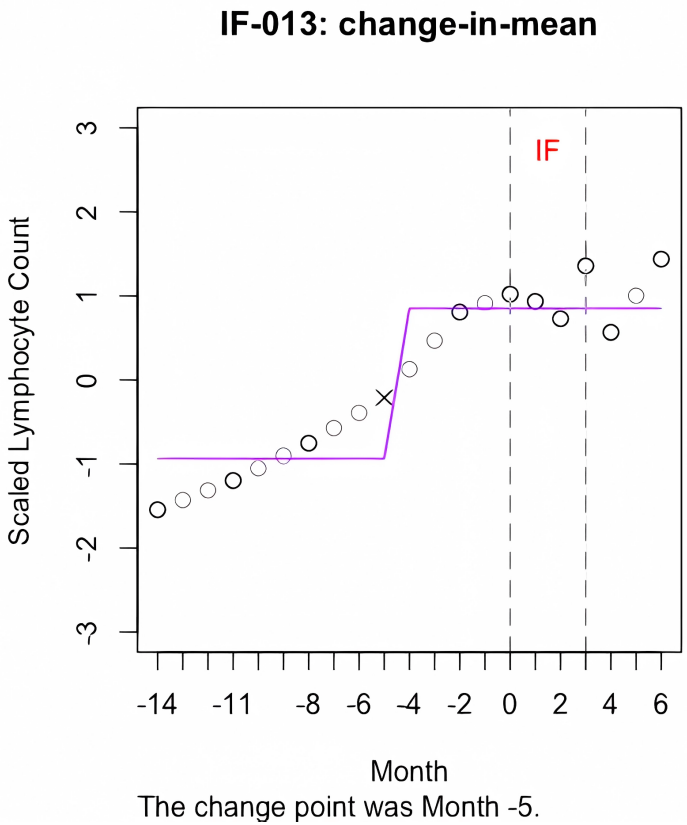
ecp – multivariate

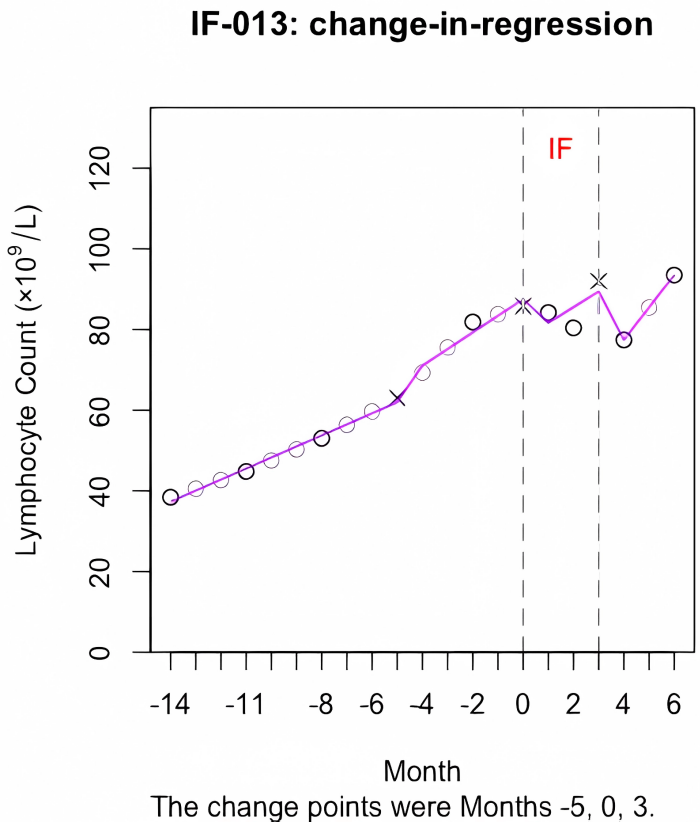
cpm – online changes

EnvCpt – changes in trend

NOT USED !

Results[2]: LYMP





(cf. *CPT.Rmd* p. 11-13 & 16)

NOT USED !

Methods[3]: LYMP

$$Y_i = f(x_i) + \epsilon_i = \beta_0 + \sum_{j=1}^{k} \beta_j B_j(x_i) + \epsilon_i \tag{2}$$

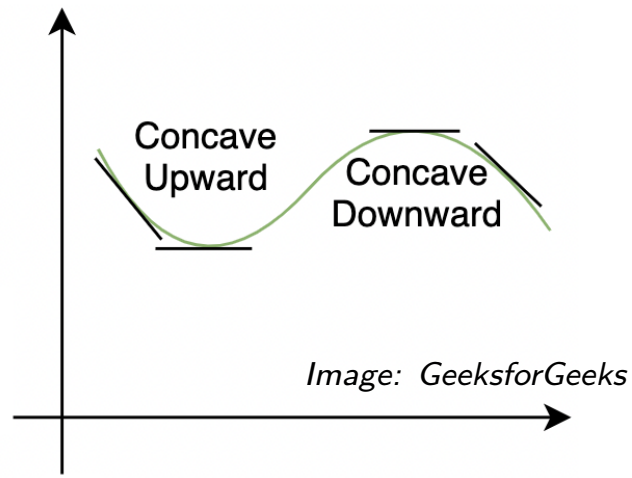
▷ A smoothing spline seeks to find a function $\hat{f}$ that minimizes

$$\sum_{i=1}^{n} \left(Y_i - \hat{f}(x_i)\right)^2 + \lambda \int \left(\hat{f}''(x)\right)^2 \mathrm{d}x \tag{3}$$

The resulting function $\hat{f}$ is a piecewise cubic polynomial.

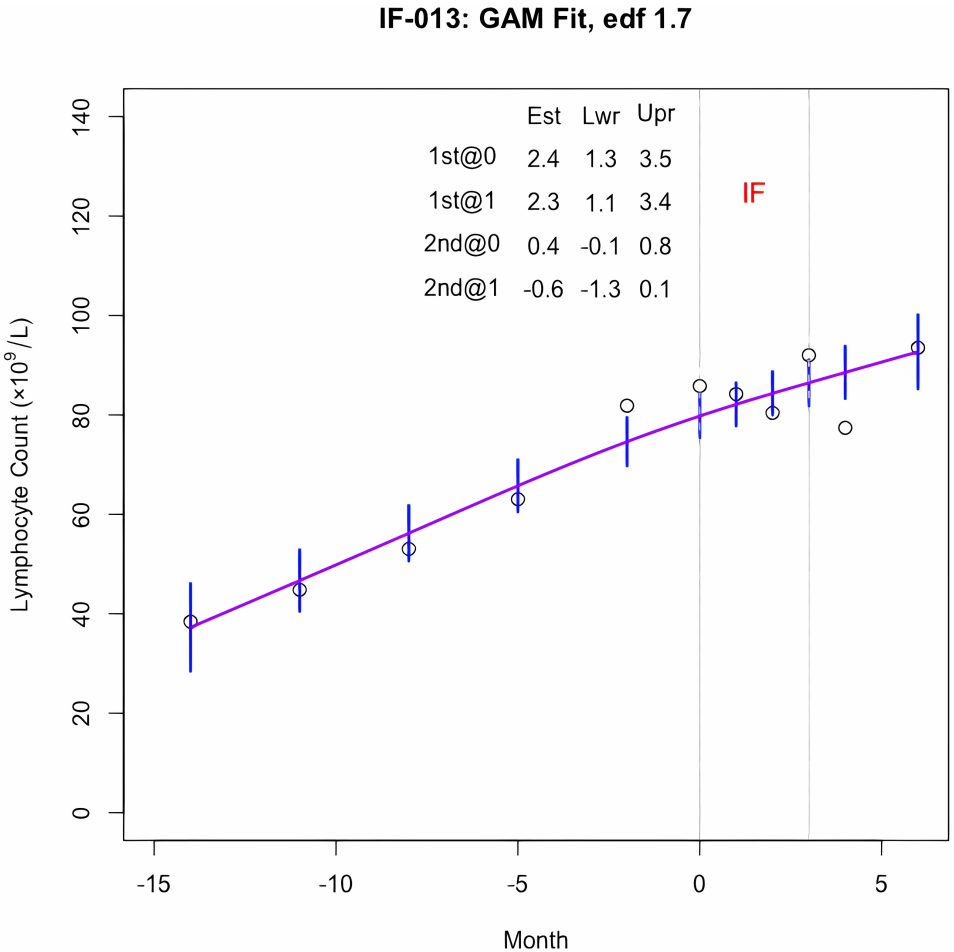
▶ A generalized additive model extends a generalized linear model to include a smooth function of the predictor such as a spline fit.

$$g(\mathbb{E}[Y_i]) = \beta_0 + s(x_i) \tag{4}$$



Derivatives:
displacement ⇀ velocity ⇀ acceleration

Results[3]: LYMP



Derivatives at Months 0 and 1.

(cf. *DER V2.1.Rmd* p. 6)

```
> df[df$ID %in% c("IF-012", "IF-018"),]
       ID        Phase                          Fast Total Global Physical Fatigue
22 IF-012      Pretest 3 Month Fast + 3 Month Post    43     58       86      11
23 IF-012 Intervention 3 Month Fast + 3 Month Post    45     66      100      22
30 IF-018      Pretest 3 Month Fast + 3 Month Post    49     58       86      33
31 IF-018 Intervention 3 Month Fast + 3 Month Post    47     75      100      33
```

Scenario 1: exclude all the subjects who missed the posttest, run omnibus RM-ANOVA for the remaining eight subjects with three conditions, check for the sphericity/circularity assumption, and may conduct the post-hoc analysis.

Scenario 2: run RM-ANOVA for all 15 subjects with two conditions (pretest and intervention).

Methods[4]: QoL

Remove the **nuisance** between-subjects variability
in within-subject designs

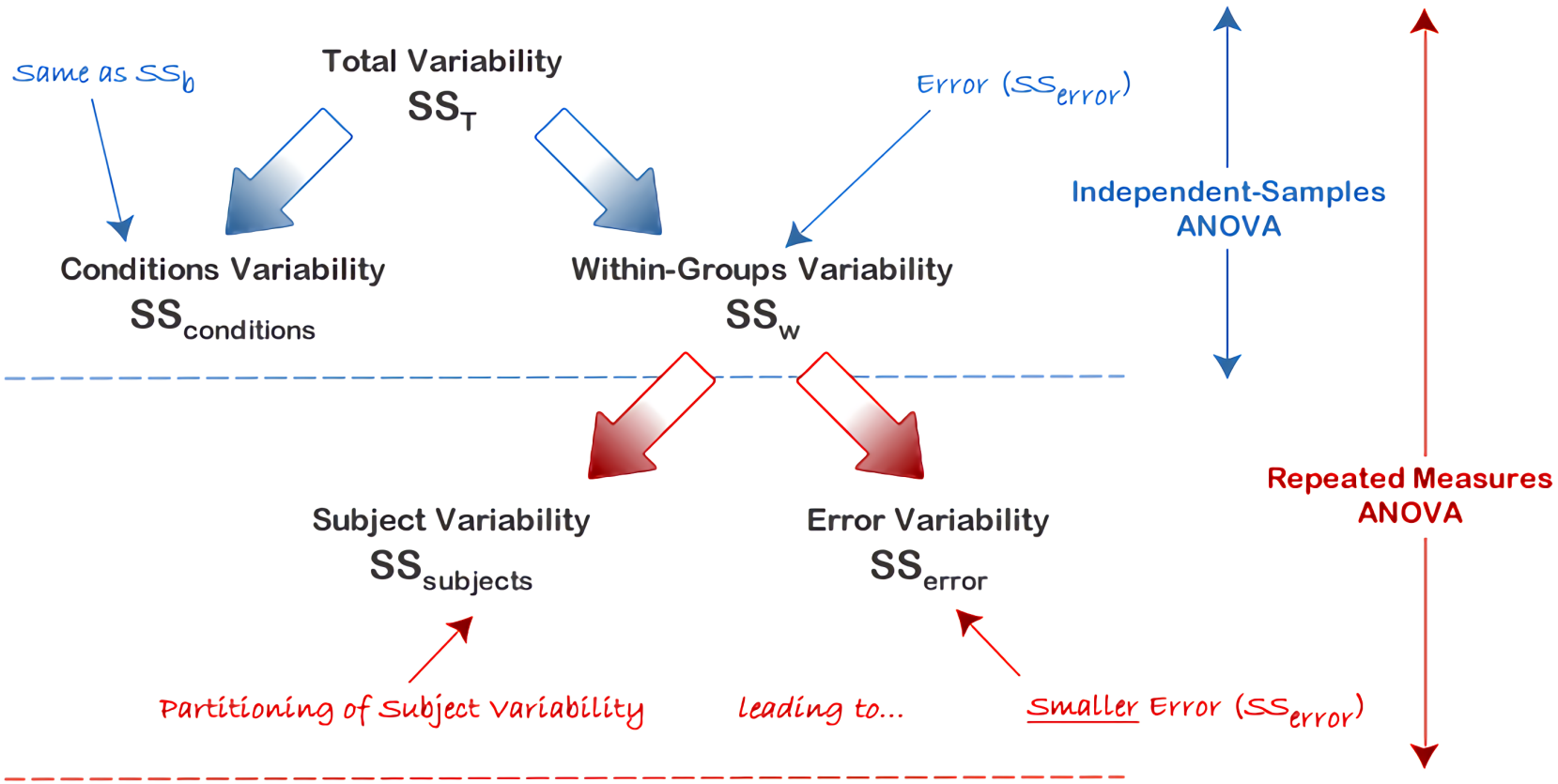


Image: Laerd Statistics

Methods[4] (continued): QoL

FURTHER READING

$$\begin{aligned} \mathcal{M}_1: \; & Y_{ij} = \mu + \sigma_\epsilon(t_i + b_j) + \epsilon_{ij} \\ \text{versus} \quad \mathcal{M}_0: \; & Y_{ij} = \mu + \sigma_\epsilon b_j + \epsilon_{ij}, \\ & \epsilon_{ij} \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma_\epsilon^2) \text{ for } i = 1, \cdots, a; \; j = 1, \cdots, n. \end{aligned} \tag{5}$$

$$\underbrace{\frac{p(\mathcal{M}_1 \mid \text{Data})}{p(\mathcal{M}_0 \mid \text{Data})}}_{\text{posterior odds}} = \underbrace{\frac{p(\text{Data} \mid \mathcal{M}_1)}{p(\text{Data} \mid \mathcal{M}_0)}}_{\text{Bayes factor } BF_{10}} \cdot \underbrace{\frac{p(\mathcal{M}_1)}{p(\mathcal{M}_0)}}_{\text{prior odds}} \tag{6}$$

- **Evidence** (supporting $\mathcal{M}_1$) is anecdotal (BF 1-3), moderate (BF 3-10), strong (BF 10-30), or very strong (BF 30-100).

$$BF_{10} = 1/BF_{01}$$

Results[4]: QoL

Global	Scenario 1	
RM-ANOVA	$p = .40$, $F(2, 14) = .97$	Not significant.
BF_{10}	0.42	Anecdotal evidence of absence.
95% Credible Intervals	Pretest:[69.46, 84.80] Intervention: [74.34, 89.68] Posttest: [69.39, 84.73]	32% separation, given $n = 8$.

Global	Scenario 2	
RM-ANOVA	$p = .022$, $F(1, 14) = 6.69$	Significant at $\alpha = .05$.
BF_{10}	2.65	Anecdotal evidence for an effect.
95% Credible Intervals	Pretest: [70.94, 80.27] Intervention: [78.03, 87.36]	76% separation, given $n = 15$.

(cf. *QoL v1.2.Rmd* p. 2-5)

```
> table(dat$ID) #number of observations for each subject

IF-002 IF-004 IF-005 IF-006 IF-007
    12     13     12     13      3

IF-009 IF-010 IF-012 IF-013 IF-014
    14     12     10     11      2

IF-017 IF-018 IF-019 IF-020 IF-021 IF-022
    25     13     23     24     21     25
```

Results

A one unit increase in time (a week) is associated with an average weight loss of 0.39 lbs (95% confidence interval 0.15-0.62 lbs), according to a longitudinal study of 14 subjects over a minimum of 12 weeks and a maximum of 26 weeks from 2021 to 2022.

When examining the individual fit, IF-010, on the contrary, was gaining weight during the intervention. (cf. *Weight v1.2.Rmd* p. 2-3)

References

Dobson, A. J., & Barnett, A. G. (2018). *An introduction to generalized linear models* (4th ed.). CRC Press. https://doi.org/10.1201/9781315182780

Wei, Z., Nathoo, F. S., & Masson, M. E. J. (2022). *rmBayes: Performing Bayesian inference for repeated-measures designs*. R package version 0.1.15. https://cran.r-project.org/package=rmBayes

Wei, Z., Nathoo, F. S., & Masson, M. E. J. (2023). Investigating the relationship between the Bayes factor and the separation of credible intervals. *Psychonomic Bulletin & Review*. https://doi.org/10.3758/s13423-023-02295-1

Thank you!
Intermittent Fasting Oncology Studies

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